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United States Patent	12385970
Kind Code	B2
Date of Patent	August 12, 2025
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Test socket and apparatus for testing a semiconductor package

Abstract

A test socket and a test apparatus for testing a semiconductor package are provided. The test socket includes: a first connection structure including a first insulating body and a plurality of conductive plugs within the first insulating body; and a second connection structure disposed on the first connection structure and including a second insulating body and a plurality of elastic conductive pillars, the second insulating body being elastic, and the plurality of elastic conductive pillars being formed by arranging a plurality of conductive particles in the second insulating body in a vertical direction; wherein the plurality of elastic conductive pillars are in vertical alignment with the plurality of plugs of the first connection structure, respectively, and the conductive particles in each of the plurality of elastic conductive pillars can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar.

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Family ID: 87932676

Appl. No.: 18/180103

Filed: March 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230288474 A1	Sep. 14, 2023

Foreign Application Priority Data

CN	202210238837.4	Mar. 11, 2022
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Publication Classification

Int. Cl.: G01R31/28 (20060101); G01R1/04 (20060101)

U.S. Cl.:

CPC G01R31/2896 (20130101); G01R1/0441 (20130101);

Field of Classification Search

CPC: G01R (31/2896); G01R (1/0441)

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Background/Summary

TECHNICAL FIELD

(1) The present application generally relates to semiconductor technology, and more particularly, to a test socket and an apparatus for testing a semiconductor package.

BACKGROUND OF THE INVENTION

(2) The semiconductor industry is constantly faced with complex integration challenges as consumers want their electronics to be smaller, faster and higher performance with more and more functionalities packed into a single device. With such needs, Ball Grid Array (BGA) type semiconductor packages which implement a high pin count by forming a plurality of external terminals having a ball shape at a bottom surface of a body portion thereof have been favored.

(3) Semiconductor packages that have undergone complicated processing are subjected to various types of electrical tests so as to test their characteristics and for defects thereof. To this end, a test

socket is used to electrically connect metallic wires or contact pads of a socket board (for example, a printed circuit board) mounted in test equipment and external terminals of a semiconductor package to be tested. That is, when testing a semiconductor package, the test socket serves as an interface to electrically connect the socket board of the test equipment and the semiconductor package under test.

(4) Therefore, a need exists for a high reliable test apparatus for testing a semiconductor package.

SUMMARY OF THE INVENTION

(5) An objective of the present application is to provide a high reliable test socket or test apparatus for testing a semiconductor package.

(6) According to an aspect of the present application, a test socket for testing a semiconductor package is provided. The test socket may include: a first connection structure including a first insulating body and a plurality of conductive plugs within the first insulating body; and a second connection structure disposed on the first connection structure and including a second insulating body and a plurality of elastic conductive pillars, the second insulating body being elastic, and the plurality of elastic conductive pillars being formed by arranging a plurality of conductive particles in the second insulating body in a vertical direction; wherein the plurality of elastic conductive pillars are in vertical alignment with the plurality of plugs of the first connection structure, respectively, and the conductive particles in each of the plurality of elastic conductive pillars can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar.

(7) According to another aspect of the present application, a test apparatus for testing a semiconductor package is provided. The test apparatus may include: a socket board having a plurality of contact pads; a test socket for connecting the contact pads of the socket board with external terminals of the semiconductor package, and including: a first connection structure including a first insulating body and a plurality of conductive plugs within the first insulating body; and a second connection structure disposed on the first connection structure and including a second insulating body and a plurality of elastic conductive pillars, the second insulating body being elastic, and the plurality of elastic conductive pillars being formed by arranging a plurality of conductive particles in the second insulating body in a vertical direction; wherein the plurality of elastic conductive pillars are in vertical alignment with the plurality of plugs of the first connection structure, respectively, and the conductive particles in each of the plurality of elastic conductive pillars can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar.

(8) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only, and are not restrictive of the invention. Further, the accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and together with the description, serve to explain principles of the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The drawings referenced herein form a part of the specification. Features shown in the drawing illustrate only some embodiments of the application, and not of all embodiments of the application, unless the detailed description explicitly indicates otherwise, and readers of the specification should not make implications to the contrary.

(2) FIG. 1 is a cross-sectional view of a Pogo type test socket for testing a semiconductor package.

(3) FIG. 2 is a cross-sectional view of a rubber type test socket for testing a semiconductor package.

- (4) FIG. 3 is a cross-sectional view of a test socket for testing a semiconductor package according to an embodiment of the present application.
- (5) FIG. 4A is an enlarged view of a portion of the test socket of FIG. 3 according to an embodiment of the present application.
- (6) FIG. 4B is an enlarged view of a portion of the test socket of FIG. 3 according to an embodiment of the present application.
- (7) FIG. 5 is a cross-sectional view of a test socket for testing a semiconductor package according to an embodiment of the present application.
- (8) FIG. 6A is a cross-sectional view of a test socket for testing a semiconductor package according to an embodiment of the present application.
- (9) FIG. 6B is a top view of the test socket of FIG. 6A according to an embodiment of the present application.
- (10) FIG. 7 is a cross-sectional view of a test apparatus for testing a semiconductor package according to an embodiment of the present application.
- (11) FIG. 8 is a top view of a support frame according to an embodiment of the present application.
- (12) FIG. 9 is a top view of a mesh according to an embodiment of the present application.
- (13) The same reference numbers will be used throughout the drawings to refer to the same or like parts.

DETAILED DESCRIPTION OF THE INVENTION

(14) The following detailed description of exemplary embodiments of the application refers to the accompanying drawings that form a part of the description. The drawings illustrate specific exemplary embodiments in which the application may be practiced. The detailed description, including the drawings, describes these embodiments in sufficient detail to enable those skilled in the art to practice the application. Those skilled in the art may further utilize other embodiments of the application, and make logical, mechanical, and other changes without departing from the spirit or scope of the application. Readers of the following detailed description should, therefore, not interpret the description in a limiting sense, and only the appended claims define the scope of the embodiment of the application.

(15) In this application, the use of the singular includes the plural unless specifically stated otherwise. In this application, the use of “or” means “and/or” unless stated otherwise. Furthermore, the use of the term “including” as well as other forms such as “includes” and “included” is not limiting. In addition, terms such as “element” or “component” encompass both elements and components including one unit, and elements and components that include more than one subunit, unless specifically stated otherwise. Additionally, the section headings used herein are for organizational purposes only, and are not to be construed as limiting the subject matter described.

(16) As used herein, spatially relative terms, such as “beneath”, “below”, “above”, “over”, “on”, “upper”, “lower”, “left”, “right”, “vertical”, “horizontal”, “side” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It should be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

(17) FIG. 1 is a cross-sectional view of a Pogo type test socket **100** for testing a semiconductor package.

(18) Referring to FIG. 1, the test socket **100** includes an insulating body **110** and a plurality of Pogo pins **120** within the insulating body **110**. The Pogo pins **120** are used for electrically connecting external terminals of a device under test (not shown) and contact pads of a test board (not shown),

and the main body **110** can fix and support the Pogo pins **120** to protect them from deformation and external physical impact. As shown in FIG. 1, each Pogo pin **120** includes a pipe-shaped pin body **122**, a metallic top contactor **124** coupled to a top end of the pin body **122**, a metallic bottom contactor **126** coupled to a bottom end of the pin body **122**, and a compressible coil spring **128** disposed inside the pin body **122**. The compressible coil spring **128** can contact with the top contactor **124** at its top end, and contact with the bottom contactor **126** at its bottom end. With these configurations, when the test socket **100** is used for testing a semiconductor package, the semiconductor package under test can be loaded onto the test socket **100**, with the top contactors **124** contacting with external terminals of the semiconductor package and the bottom contactors **126** contacting with contact pads of a test board. An external force can be applied onto the semiconductor package, thereby the semiconductor package pressing the top contactors **124** of the test socket **100** downward. As such, the coil spring **128** can provide an elastic connection between the external terminals of the semiconductor package and the contact pads of the test board.

(19) However, when the test socket **100** is connected to solder balls or solder bumps of the semiconductor package under test, the top contactors **124** may scratch and damage the solder balls or the solder bumps, producing defects such as burrs. The burrs may establish undesired short circuit currents with its adjacent solder balls or solder bumps, bringing down the yield.

Furthermore, due to a substantial length and the coiled shape of the springs **128**, a signal integrity (SI) problem may be caused at a high-frequency signal test.

(20) FIG. 2 is a cross-sectional view of a rubber type test socket **200** for testing a semiconductor package.

(21) Referring to FIG. 2, the test socket **200** includes an insulating body **210** and a plurality of pressure conductive rubber pillars **220** within the insulating body **210**. The insulating body **210** may be made of solidified insulating silicone powder, for example. The pressure conductive rubber pillars **220** may be made by dispersing conductive powder **225** into predetermined sites of the insulating body **210**. As shown in FIG. 2, the pressure conductive rubber pillars **220** are positioned and oriented perpendicular to the insulating body **210**. A top surface of each pressure conductive rubber pillar **220** is exposed from a top surface of the insulating body **210**, and a bottom surface of each pressure conductive rubber pillar **220** is exposed from a bottom surface of the insulating body **210**. When the test socket **200** is used for testing a semiconductor package (not shown), the semiconductor package is loaded onto the top surface of the insulating body **210**, with the top surfaces of the pressure conductive rubber pillars **220** contacting with external terminals of the semiconductor package and the bottom surfaces of the pressure conductive rubber pillars **220** contacting with contact pads of a test board (not shown). An external force can be applied onto the semiconductor package under test, thereby the semiconductor package pressing against the top surfaces of the pressure conductive rubber pillars **220**. In this way, the conductive powder **225** within the pressure conductive rubber pillars **220** can come into contact with each other, so as to produce electrical conductivity between the external terminals of the semiconductor package and the contact pads of the test board.

(22) However, the rubber type test socket **200** as shown in FIG. 2 has a higher resistance than the Pogo type test socket **100** as shown in FIG. 1, and thus cannot bear large current consumption.

(23) To address at least one of the above problems, a test socket for testing a semiconductor package is provided. The test socket may include a first connection structure and a second connection structure disposed on the first connection structure. The first connection structure may include a first insulating body and a plurality of conductive plugs within the first insulating body. The second connection structure may be a rubber type test socket, for example, similar as the rubber type test socket **200** shown in FIG. 2, but may have a shorter length. Specifically, the second connection structure may include a second insulating body and a plurality of elastic conductive pillars. The plurality of elastic conductive pillars may be formed by dispersing a plurality of conductive particles into an elastic material, and the pillars may extend in a thickness direction of

the second insulating body and passes through the second insulating body. The plurality of elastic conductive pillars may be in vertical alignment with the plurality of plugs of the first connection structure, respectively. In response to an external pressure applied onto each elastic conductive pillar, the conductive particles in the elastic conductive pillar can produce electrical conductivity. Compared with a conventional Pogo type test socket, the test socket according to embodiments of the present application can provide a wider contact area and better high-frequency performance, and would not damage external terminals of the semiconductor package under test. Compared with a conventional rubber type test socket, the test socket according to the embodiments of the present application can bear more current consumption because a large portion of the test socket contains conductive plugs having high electrical conductivities.

(24) Referring to FIG. 3, a cross-sectional view of a test socket **300** is illustrated according to an embodiment of the present application. As shown in FIG. 3, the test socket **300** may include a first connection structure **310** and a second connection structure **320**.

(25) The first connection structure **310** may include a first insulating body **312** and a plurality of conductive plugs **314** within the first insulating body **312**. The first insulating body **312** can fix and support the conductive plugs **314** to protect them from deformation and external physical impact. In some embodiments, the first insulating body **312** may be made of a plastic material or a resin material, for example, a thermoset resin. In some embodiments, the conductive plugs **314** may be made of a metallic material having excellent electrical conductivity, such as Al, Cu, Ag, Pt, Au, and the like. The present application does not limit the material of the first insulating body **312** and the conductive plugs **314** to that disclosed herein.

(26) In some embodiments, the conductive plug **314** may be a Pogo pin without spring. That is, the plug **314** may include a pipe-shaped pin body, a metallic top contactor coupled to a top end of the pin body, and a metallic bottom contactor coupled to a bottom end of the pin body. The top contactor may contact with the bottom contactor. Thus, the plug **314** can be manufactured using a process for manufacturing a Pogo pin without forming the spring between the top contactor and the bottom contactor.

(27) In some embodiments, the plug **314** is formed as a single piece, and can be formed in a single step. For example, a plurality of through-holes can be formed through the first insulating body **312**, and then a metallic material can be deposited or plated into the through-holes to form the plugs **314**.

(28) As shown in FIG. 3, a bottom surface **314b** of the conductive plug **314** may protrude from a bottom surface **312b** of the first insulating body **312**, so as to facilitate the contact with an external contact pad or a metallic wire of a test board (not shown). In some other embodiments, the bottom surface **314b** of the conductive plug **314** may be coplanar with the bottom surface of the first insulating body **312**.

(29) FIG. 4A is an enlarged view of a portion **330** of the test socket **300** of FIG. 3 according to an embodiment of the present application. As shown in FIG. 4A, a protruding portion **315** is formed on the top surface **314a** of the conductive plug **314** and embedded into an elastic conductive pillar **324** of the second connection structure. The protruding portion **315** can provide a larger interface between the elastic conductive pillar **324** and the conductive plug **314**, and thus the adhesion and conductivity between the conductive plug **314** and the elastic conductive pillar **324** can be improved significantly. In the example shown in FIG. 4A, the protruding portion **315** has a rectangular cross section. In other examples, the protruding portion **315** may also have a trapezoidal, triangular, or semicircular cross section.

(30) FIG. 4B is an enlarged view of a portion **330** of the test socket **300** of FIG. 3 according to another embodiment of the present application. As shown in FIG. 4B, more than one protruding portions **316** are formed on the top surface of the conductive plug **314**, which can further increase the contact area at the interface between the elastic conductive pillar **324** and the conductive plug **314**.

(31) However, the present application does not limit the specific number or shape of the protruding portion(s) formed on the top surface of the conductive plug to those disclosed in FIG. 4A and FIG. 4B, and even in some other embodiments, there may be no protruding portion formed on the top surface of the conductive plug, that is, the top surface of the conductive plug is flat.

(32) Still referring to FIG. 3, the second connection structure 320 is disposed on and connected to the first connection structure 310. The second connection structure 320 includes a second insulating body 322 and a plurality of elastic conductive pillars 324 inside the second insulating body 322.

(33) The plurality of elastic conductive pillars 324 may be formed by dispersing a plurality of conductive particles 325 into an elastic material, and the pillars 324 may extend in a thickness direction of the second insulating body 322. In an example, insulating powder and conductive powder may be mixed at a predefined ratio and melted in a mold frame. The insulating powder may be made of non-conductive silicone such as silicon rubber, and the conductive powder may be metallic powder such as gold (Au) or nickel (Ni). Then, the conductive powder can be brought to a site where the elastic conductive pillars 324 will be formed. Specifically, a positive voltage and the ground or a negative voltage may be applied to the two opposite surfaces of the second insulating body 322, respectively. The voltage difference may then cause the conductive particles 325 contained in the melted mixture to migrate into a columnar region so as to form the elastic conductive pillars 324. Afterwards, the second connection structure 320 is completed by solidifying the melted mixture.

(34) As shown in FIG. 3, the plurality of elastic conductive pillars 324 are in vertical alignment with the plurality of plugs 314 of the first connection structure 310, respectively, and the conductive particles 325 in each of the plurality of elastic conductive pillars 324 can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar 324.

(35) In some embodiments, a top surface 324a of the elastic conductive pillar 324 may protrude from a top surface 322a of the second insulating body 322, so as to facilitate the pillar 324 contacting with an external terminal of a semiconductor package under test.

(36) In some embodiments, a thickness of the first insulating body is 0.5 to 5 times (for example, 1, 2, 3 or 4 times) a thickness of the second insulating body. That is, a large portion of the test socket is composed of the first connection structure with the conductive plugs having high electrical conductivity. Thus, the test socket of the embodiments of the present application can bear more current consumption than a conventional rubber type test socket such as the rubber type test socket 200 shown in FIG. 2.

(37) Referring to FIG. 5, a cross-sectional view of a test socket 500 is illustrated according to another embodiment of the present application. As shown in FIG. 5, the test socket 500 may include a first connection structure 510 and a second connection structure 520, which are similar as the first connection structure 310 and the second connection structure 320 of the test socket 300 shown in FIG. 3, and will not be elaborated herein. Different from the test socket 300 shown in FIG. 3, the test socket 500 shown in FIG. 5 further includes a plurality of contact pads 550 which are disposed on respective top surfaces 524a of a plurality of elastic conductive pillars 524.

(38) In particular, the elastic conductive pillars 524 are relatively soft and thus are susceptible to wear or damage caused by repeated contact with the external terminals (for example, solder balls) of a semiconductor package under test. Defects may develop from hollows or voids or pits at the top surfaces of the elastic conductive pillars caused by the repeated pressure from the solder balls of the semiconductor package under test. In order to solve the problem, the contact pads 550 are formed on the top surfaces 524a of the elastic conductive pillars 524.

(39) In some embodiments, the contact pad 550 may be a metal plate. The metal plate may have a higher stiffness, such that the metal plate can withstand higher contact pressure. The metal plate may also have a better conductivity than the elastic conductive pillar 524, such that the metal plate can form a robust electric connection with the solder ball of the semiconductor package under test,

even without excessive contact pressure by the semiconductor package. Thus, the contact pads **550** can protect the elastic conductive pillars **524**, and thereby helping lengthen the life span of the test socket **500**.

(40) In some embodiments, the contact pad **550** may be a metal ring. For example, a top end of the elastic conductive pillar **524** may protrude from a top surface **522a** of the second insulating body **522**, and the metal ring may be disposed at a periphery of the top end of the elastic conductive pillar **524**. The metal ring can similarly protect the elastic conductive pillar **524** and help lengthen the life span of the test socket **500**.

(41) According to an embodiment of the present application, a cross-sectional view and a top view of a test socket **600** are respectively illustrated in FIGS. **6A** and **6B**. FIG. **6A** is a cross-sectional view of the test socket **600** along a section line A-A' shown in FIG. **6B**.

(42) As shown in FIG. **6A**, the test socket **600** may include a first connection structure **610** and a second connection structure **620**, which are similar as the first connection structure **310** and a second connection structure **320** of the test socket **300** shown in FIG. **3**, and will not be elaborated herein. Different from the test socket **300** shown in FIG. **3**, the test socket **600** shown in FIGS. **6A** and **6B** further includes one or more Pogo pins **660**. In particular, the Pogo pin **660** may extend through a first insulating body **612** of a first connection structure **610** and the second insulating body **622** of the second connection structure **620**.

(43) Referring to FIG. **6**, the Pogo pin **660** includes a pipe-shaped pin body **662**, a metallic top contactor **664** coupled to a top end of the pin body **662**, a metallic bottom contactor **666** coupled to a bottom end of the pin body **662**, and a compressible coil spring **668** disposed inside the pin body **662**. The compressible coil spring **668** contacts with the top contactor **664** at its top end, and contacts with the bottom contactor **666** at its bottom end. In an example, the pipe-shaped pin body **662**, the top contactor **664** and the bottom contactor **666** are made of brass or copper, and the coil spring **668** is made of copper alloys or spring steel. However, the present application does not limit the material of the Pogo pin **660** to that disclosed herein. With these configurations, when the test socket **600** is used for testing a semiconductor package, the semiconductor package can be loaded onto the test socket **600**, with the top contactors **664** contacting with external terminals of the semiconductor package and the bottom contactors **666** contacting with contact pads of a test board. Furthermore, an external force can be applied onto the semiconductor package under test, such that the external terminals of the semiconductor package can press against the top contactors **664**, and the coil spring **668** can provide an elastic contact between the external terminals of the semiconductor package and the contact pads of the test board.

(44) In the example shown in FIG. **6**, the bottom surfaces **666b** of the bottom contactors **666** may protrude from the bottom surface **612b** of the first insulating body **612** so as to facilitate a contact with the contact pads of the test board. The top surfaces **664a** of the top contactors **664** may protrude from a top surface **622a** of a second insulating layer **622**. The top surfaces **664a** may have peaks and valleys so as to facilitate contacting with external terminals of a semiconductor package under test.

(45) The Pogo pin **660** of the test socket **600** may have a lower resistance and higher current capacity than the pins composed of the plug **614** and the elastic conductive pillar **624**. Therefore, the Pogo pin **660** can be used to connect a high current consumption terminal of the semiconductor package under test with the test board. It could be understood that, the number or location of the Pogo pin **660** shown in FIGS. **6A** and **6B** is exemplary only, and may vary according to actual needs.

(46) According to another aspect of the present application, a test apparatus including any of the test sockets described above is also provided.

(47) Referring to FIG. **7**, a cross-sectional view of a test apparatus is illustrated according to an embodiment of the present application. As shown in FIG. **7**, the test apparatus may include a test socket **700** and a socket board **760**.

(48) The test socket **700** shown in FIG. 7 is similar to the test socket **300** shown in FIG. 3. Specifically, the test socket **700** includes a first connection structure **710** and a second connection structure **720** disposed on the first connection structure **710**. The first connection structure **710** includes a first insulating body **712** and a plurality of conductive plugs **714** within the first insulating body **712**. The second connection structure **720** includes a second insulating body **722** and a plurality of elastic conductive pillars **724** within the second insulating body **722**. The plurality of elastic conductive pillars **724** may be formed by dispersing a plurality of conductive particles into an elastic material in a thickness direction of the second insulating body **722**. The plurality of elastic conductive pillars **724** are in vertical alignment with the plurality of plugs **714** of the first connection structure **710**, respectively, and the conductive particles in each of the plurality of elastic conductive pillars **724** can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar **724**. More detail about the test socket **700** may refer to the test sockets described with reference to FIGS. 3-6, and will not be elaborated herein.

(49) The socket board **760** may include a plurality of contact pads **762**. The contact pads **762** may be coupled with a signal generator via various connection means, for example, traces, plugs or redistribution structures (RDSs) within the socket board. The signal generator can generate different signals for testing the semiconductor package. The test socket **700** can be mounted onto the socket board **760** to connect the contact pads **762** of the socket board **760** with external terminals **772** of the semiconductor package **770** under test. More details about the mounting will be described below with reference to FIG. 8 and FIG. 9.

(50) In some embodiments, referring to FIG. 7 and FIG. 8, the test apparatus may further include a support frame **780**. The support frame **780** may have a plane shape and be made of metal or plastics. The support frame **780** may have an opening in the center, and the test socket **700** may be fixed within the opening of the support frame **780**. The support frame **780** may have alignment holes **782** for alignment means at its four corners. The support frame **780** can be assembled with the socket board **760** via the alignment means, and thus the test socket **700** can be mounted onto the socket board **760**. For example, the assembling can be implemented by inserting a screw **775** as an alignment means into each of the alignment holes **782** of the support frame **780** and fixing the screw **775** therein. Accordingly, the screws **775** can removably fix the support frame **780** with the socket board **760**.

(51) In some embodiments, as shown in FIG. 7 and FIG. 9, the test apparatus may further include a mesh **790**. The mesh **790** may be attached to the support frame **780** as shown in FIG. 7. The mesh **790** has a plurality of holes **794** in which external terminals (for example, solder balls) of the semiconductor package may sit. For example, the holes **794** may be constructed at the same interval as that of the elastic conductive pillars **724** of the test socket shown in FIG. 7. Accordingly, the mesh **790** can determine the sitting location of solder balls of the semiconductor package under test and prevent variation of the solder balls in position when the semiconductor package is being tested. The mesh **790** has four alignment holes **792**, in which screws are inserted for aligning the mesh **790** with the support frame **780** and the socket board **760**.

(52) Still referring to FIG. 7, a process for testing a semiconductor package **770** using the test apparatus of the present application will be described below.

(53) First, the socket board **760** having the test socket **700** installed thereon is prepared. The bottom surfaces of the elastic conductive pillars **724**, exposed from the bottom surface of the second insulating body **722**, contact with respective contact pads of the socket board **760** through the plurality of conductive plugs **714** within the first insulating body **712** to form electrical connection therebetween.

(54) Second, the semiconductor package **770** is aligned on the test socket **700** so that solder balls **772** of the semiconductor package **770** face top parts of the elastic conductive pillars **724**. Then, the semiconductor package is pressed downwards (as indicated by a dashed arrow in FIG. 7) to closely contact with the test socket **700**. As such, the solder balls **772** of the semiconductor package **770**

can elastically contact with the elastic conductive pillars 724. Specifically, the downward force applied on the semiconductor package 770 (or, alternatively but still within the spirit and scope of the invention, upward on the test socket 700) causes the hemispheric solder balls 772 to slightly depress the top surfaces of the elastic conductive pillars 724 and to be compressively cradled. The depressed elastic conductive pillars 724 recess downwardly and inwardly to cradle bottom peripheries of the solder balls 772, thereby forming smooth but forceful and slightly expanded contact regions (which may be semicircular rather than linear). The force applied by the semiconductor package 770 can be absorbed by the elastic conductive pillars 724 and the second insulating body 722, so that the solder balls 772 of the semiconductor package 770 elastically contact with the elastic conductive pillars 724, forming a robust and reliable electrical connection. After establishing an electrical connection between the socket board 760 and the semiconductor package 770 via the test socket 700, one or more test processes may be performed by supplying test signals from the contact pad 762 of the socket board 760 through the plugs 714 and the elastic conductive pillars 724 to the solder balls 772 of the semiconductor package 770. In this way, signal paths within the semiconductor package 770 can be tested.

(55) The discussion herein included numerous illustrative figures that showed various portions of a test socket and a test apparatus. For illustrative clarity, such figures did not show all aspects of each example assembly. Any of the example assemblies and/or methods provided herein may share any or all characteristics with any or all other assemblies and/or methods provided herein.

(56) Various embodiments have been described herein with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. Further, other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of one or more embodiments of the invention disclosed herein. It is intended, therefore, that this application and the examples herein be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following listing of exemplary claims.

Claims

1. A test socket for testing a semiconductor package, comprising: a first connection structure comprising a first insulating body and a plurality of conductive plugs within the first insulating body, wherein the conductive plugs are made of a metallic material; a second connection structure disposed on the first connection structure and comprising a second insulating body and a plurality of elastic conductive pillars, the second insulating body being elastic, and the plurality of elastic conductive pillars being formed by arranging a plurality of conductive particles in the second insulating body in a vertical direction; wherein the plurality of elastic conductive pillars are in vertical alignment with the plurality of conductive plugs of the first connection structure, respectively, and the conductive particles in each of the plurality of elastic conductive pillars can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar; and one or more Pogo pins extending through the first insulating body and the second insulating body, wherein the one or more Pogo pins have a lower resistance than pins composed of the conductive plugs and the elastic conductive pillars, and are configured to connect with a high current consumption terminal of the semiconductor package under test.
2. The test socket of claim 1, wherein a bottom surface of the conductive plug protrudes from a bottom surface of the first insulating body.
3. The test socket of claim 1, wherein one or more protruding portions are formed on a top surface of the conductive plug.
4. The test socket of claim 1, wherein a top surface of the elastic conductive pillar protrudes from a top surface of the second insulating body.

5. The test socket of claim 1, wherein a contact pad is formed on a top surface of the elastic conductive pillar.
 6. The test socket of claim 1, wherein a thickness of the first insulating body is 0.5 to 5 times a thickness of the second insulating body.
 7. The test socket of claim 1, wherein the conductive particles comprise gold or nickel.
 8. A test apparatus for testing a semiconductor package, comprising: a socket board having a plurality of contact pads; and a test socket for connecting the contact pads of the socket board with external terminals of the semiconductor package, and comprising: a first connection structure comprising a first insulating body and a plurality of conductive plugs within the first insulating body, wherein the conductive plugs are made of a metallic material; a second connection structure disposed on the first connection structure and comprising a second insulating body and a plurality of elastic conductive pillars, the second insulating body being elastic, and the plurality of elastic conductive pillars being formed by arranging a plurality of conductive particles in the second insulating body in a vertical direction; wherein the plurality of elastic conductive pillars are in vertical alignment with the plurality of plugs of the first connection structure, respectively, and the conductive particles in each of the plurality of elastic conductive pillars can produce electrical conductivity in response to an external pressure applied onto the elastic conductive pillar; and one or more Pogo pins extending through the first insulating body and the second insulating body, wherein the one or more Pogo pins have a lower resistance than pins composed of the conductive plugs and the elastic conductive pillars, and are configured to connect with a high current consumption terminal of the semiconductor package under test.
 9. The test apparatus of claim 8, further comprising: a support frame for mounting the test socket onto the socket board.
 10. The test apparatus of claim 8, further comprising: a mesh positioned between the semiconductor package and the test socket and including a plurality holes, which are configured to receive the external terminals of the semiconductor package such that the external terminals are able to contact the elastic conductive pillars through the holes.
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